

추세 추종 전략

Managed Futures 는 무엇인가?

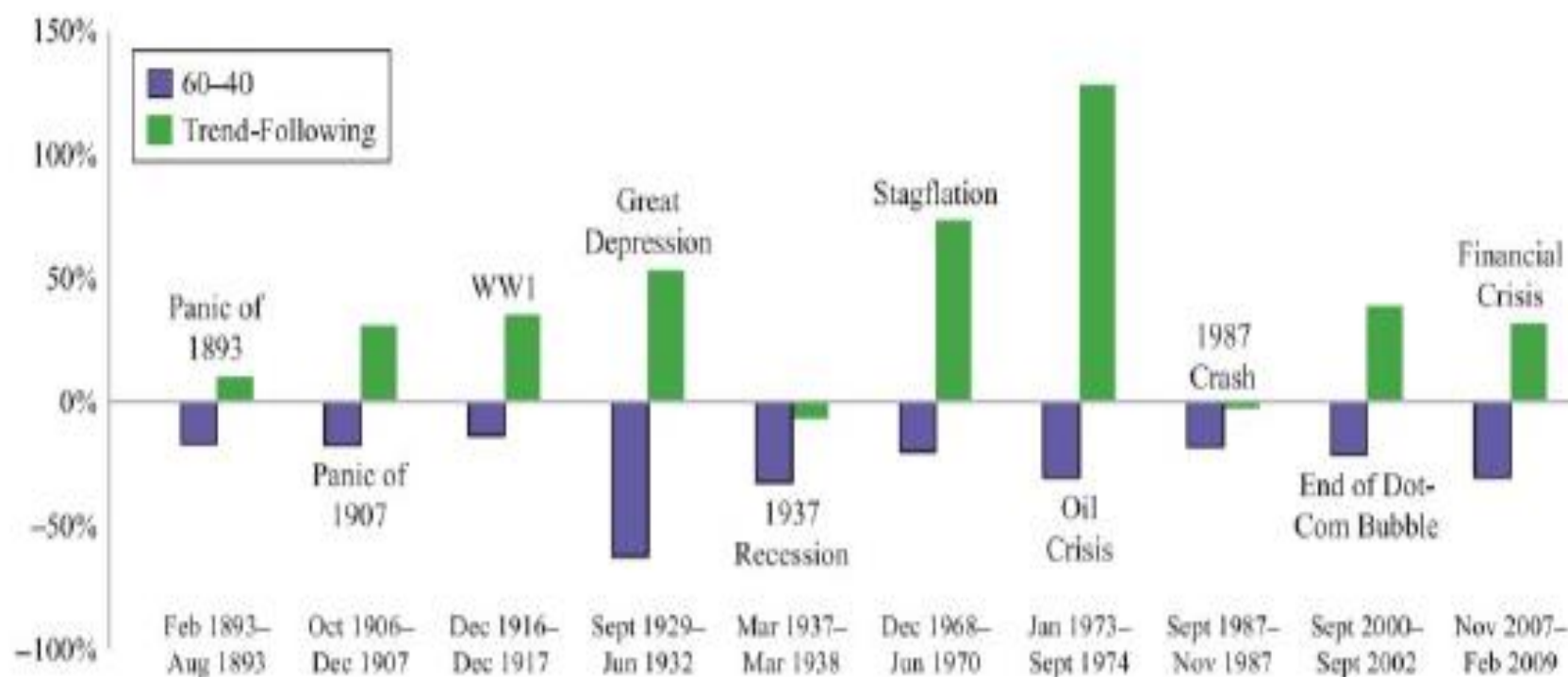
- CTAs (선물 거래 자문사)와 헤지펀드는 최소한 1970년대부터 관리된 선물 전략을 추구해왔다.
- 유동성이 높은 거래소 거래 선물 및 선물 관련 상품을 사용하여 글로벌 시장의 방향에 대한 견해를 투자한다.
 - 상품, 주식, 고정 수익 선물 및 통화 선물
- 대부분의 관리 선물(매니지드 퓨처스) 전략의 주요 동인은 추세 추종 또는 시계열 모멘텀 투자이다.
- 간단히 말해, 시계열 모멘텀 투자는
 - 상승하는 자산을 매수하고
 - 하락하는 자산을 매도하는 것이다.

Managed Futures 가 왜 지금 주목을 받고 있는가?

- Hurst, Ooi, and Pedersen (2017, JPM).

EXHIBIT 6

Time-Series Momentum during the 10 Worst Drawdowns for 60/40



Note: This exhibit shows the returns of time-series momentum (gross of fee, net of cost) and return of the portfolio that invested 60% in U.S. stocks and 40% in U.S. bonds over the 10 time periods selected as the largest drawdowns for the latter portfolio.

추세의 라이프 사이클

□ Pedersen (2015)

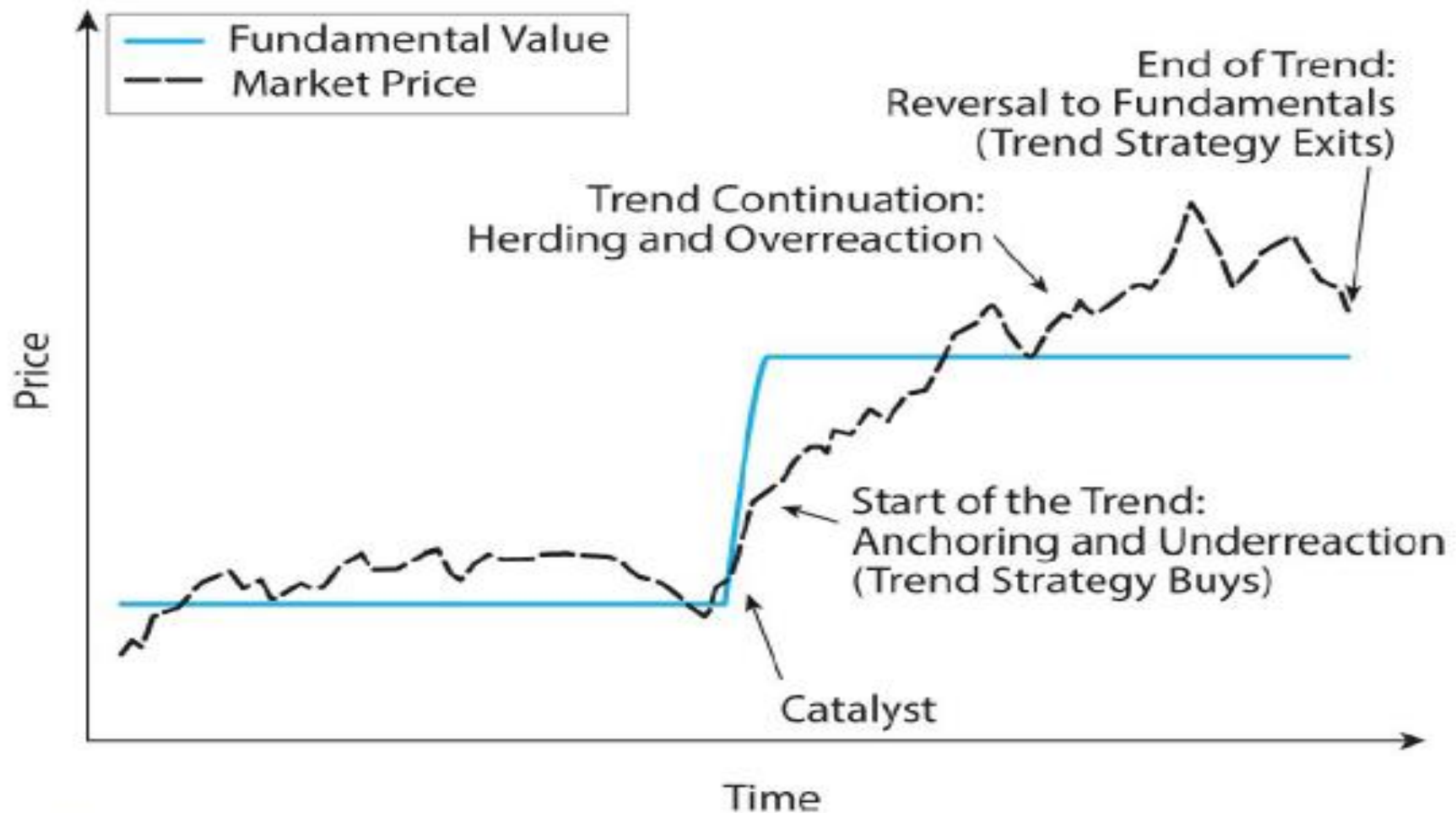


Figure 12.1. Stylized plot of the life cycle of a trend.
Source: Hurst, Ooi, and Pedersen (2013).

시계열 모멘텀의 증거

- Moskowitz, Ooi and Pedersen (2012, JFE)

$$\frac{r_t^s}{\sigma_{t-1}^s} = \alpha + \beta_h \text{sign}(r_{t-h}^s) + \epsilon_t^s$$

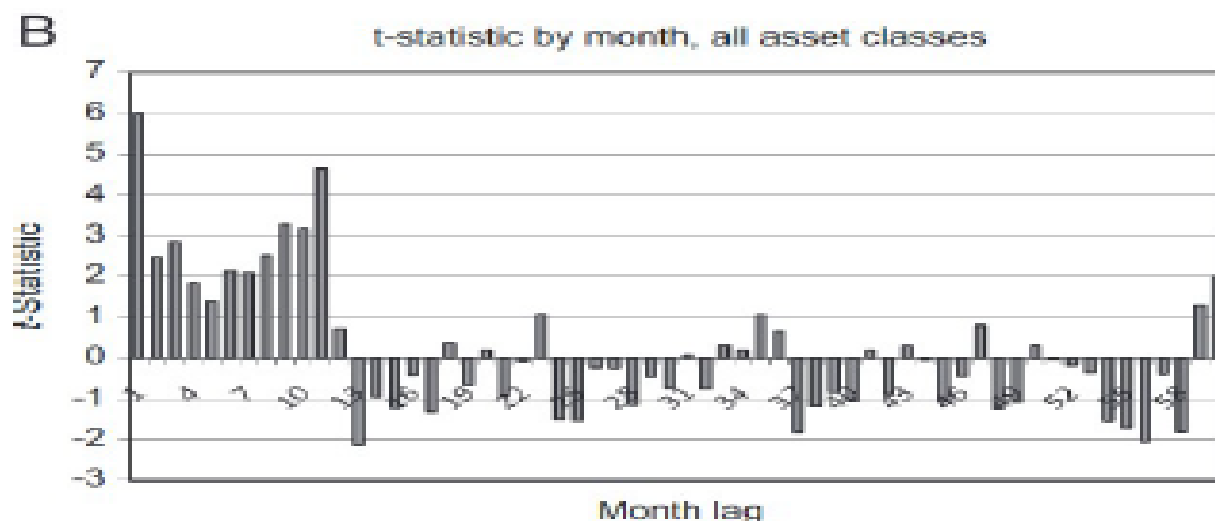


Fig. 1. Time series predictability across all asset classes. We regress the monthly excess return of each contract on its own lagged excess return over various horizons. Panel A uses the size of the lagged excess return as a predictor, where returns are scaled by their ex ante volatility to make them comparable across assets, Panel B uses the sign of the lagged excess return as a predictor, where the dependent variable is scaled by its ex ante volatility to make the regression coefficients comparable across different assets, and Panel C reports the results of the sign regression by asset class. Reported are the pooled regression estimates across all instruments with t -statistics computed using standard errors that are clustered by time (month). Sample period is January 1985 to December 2009. (A) Panel A: $r_t^s / \sigma_{t-1}^s = \alpha + \beta_h r_{t-h}^s / \sigma_{t-h-1}^s + \epsilon_t^s$; (B) Panel B: $r_t^s / \sigma_{t-1}^s = \alpha + \beta_h \text{sign}(r_{t-h}^s) + \epsilon_t^s$; (C) Panel C: Results by asset class.

추세 거래: 간단한 관리 선물 전략

- 추세에 따라 거래하는 간단한 방법을 고려하자: 실제 CTAs는 종종 더 정교한 방법을 사용한다.
- 유니버스: 가장 유동성이 높은 24개의 상품 선물, 9개의 주가지수 선물, 15개의 채권 선물 및 12개의 통화 선물
- 추세 방향: 지난 12개월 동안 초과 수익률이 양수면 롱 포지션을 취하고, 음수면 숏 포지션을 취한다.
- 포지션 크기: 각 상품에 대해 연간 변동성을 40%로 설정하여 포지션을 유지한다. (자산 s)

$$r_{t,t+1}^{TSMOM,s} = \text{sign}(r_{t-12,t}^s) \frac{40\%}{\sigma_t^s} r_{t,t+1}^s$$

- 재조정: 월간/주간/일간

간단한 관리 선물 전략의 백테스트: 12개월

□ Moskowitz, Ooi and Pedersen (2012, JFE)

$$r_{t,t+1}^{TSMOM} = \frac{1}{S_t} \sum_{s=1}^{S_t} \text{sign}(r_{t-12,t}^s) \frac{40\%}{\sigma_t^2} r_{t,t+1}^s$$

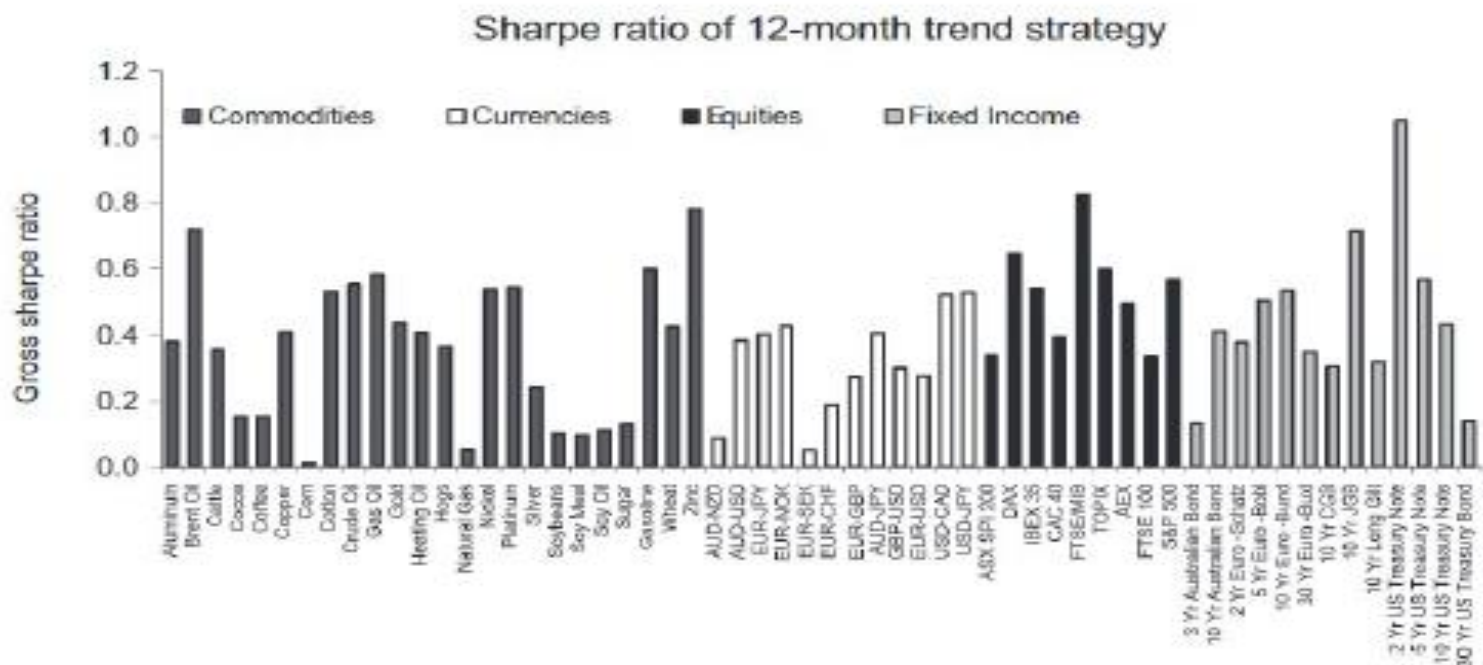


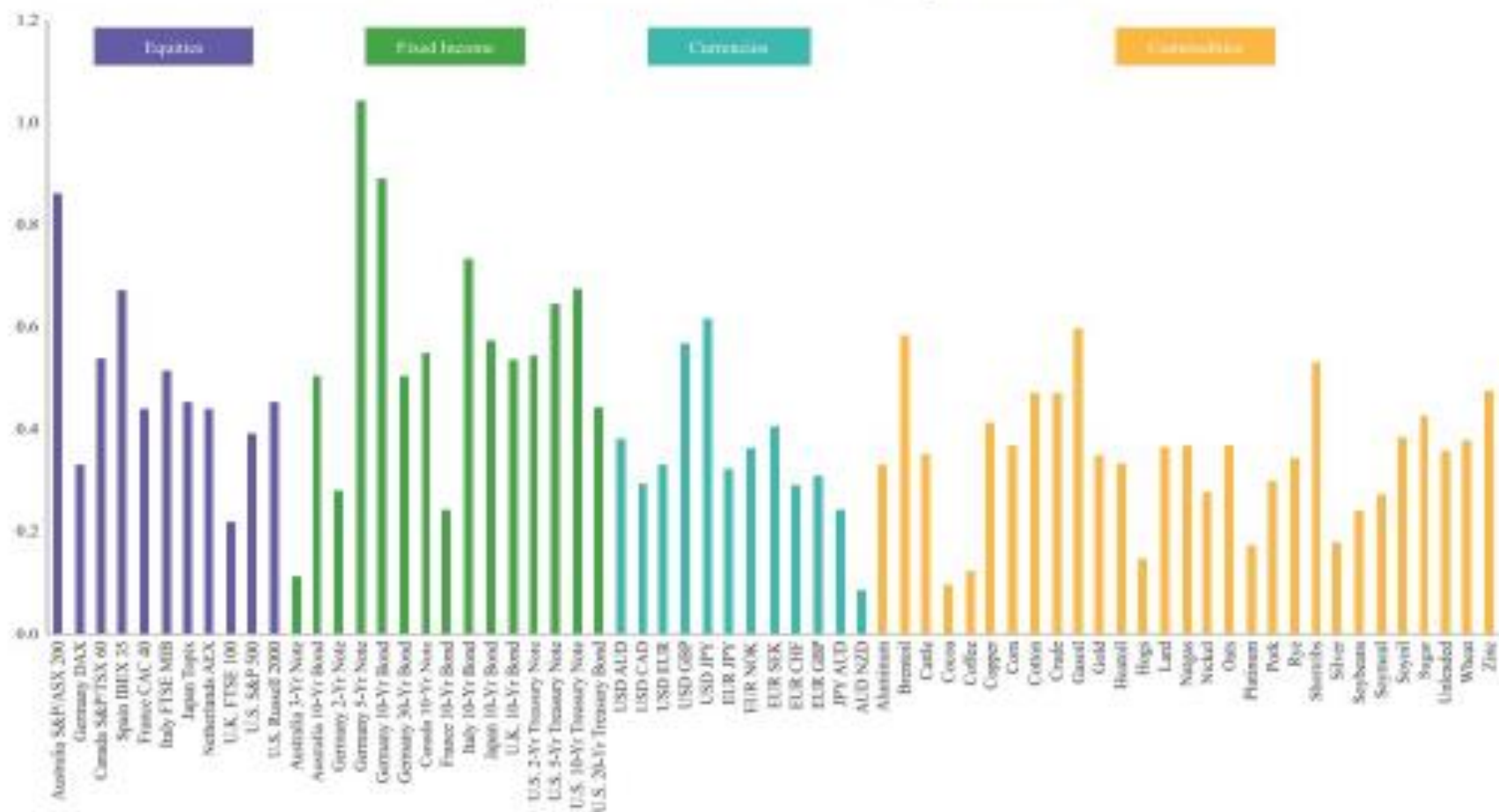
Fig. 2. Sharpe ratio of 12-month time series momentum by instrument. Reported are the annualized gross Sharpe ratio of the 12-month time series momentum or trend strategy for each futures contract/instrument. For each instrument in every month, the trend strategy goes long (short) the contract if the excess return over the past 12 months of being long the instrument is positive (negative), and scales the size of the bet to be inversely proportional to the ex ante volatility of the instrument to maintain constant volatility over the entire sample period from January 1985 to December 2009. The second figure plots a normalized value of the illiquidity of each futures contract measured by ranking contracts within each asset class by their daily trading volume (from highest to lowest) and reordering the standard normalized rank for each contract within each asset class. Positive (negative) values imply

간단한 관리선물 전략의 백테스트: 100년의 증거

□ Hurst, Ooi and Pedersen (2017, JPM)

EXHIBIT 3

Time-Series Momentum Performance by Individual Asset: Full Sample, 1880–2016



Note: This exhibit shows the Sharpe ratio of time-series momentum (gross of fee, gross of cost) by asset.

간단한 관리선물 전략의 백테스트: 100년의 증거

□ Hurst, Ooi and Pedersen (2017, JPM)

EXHIBIT 2

Performance of Time-Series Momentum by Signal

Time Period	1-Month Strategy	1-Month Strategy (lagged)	3-Month Strategy	3-Month Strategy (lagged)	12-Month Strategy	12-Month Strategy (lagged)
Full Sample						
Jan 1880–Dec 2016	1.38	0.45	1.19	0.64	1.32	1.04
By Decade						
Jan 1880–Dec 1889	1.02	−0.34	0.78	−0.23	1.03	0.89
Jan 1890–Dec 1899	1.32	0.34	0.91	0.25	1.35	0.83
Jan 1900–Dec 1909	0.87	0.35	1.20	0.65	1.52	1.52
Jan 1910–Dec 1919	0.80	0.01	0.63	0.39	0.99	0.83
Jan 1920–Dec 1929	1.75	0.56	1.23	0.51	1.77	1.27
Jan 1930–Dec 1939	1.19	0.27	1.21	0.29	1.09	0.70
Jan 1940–Dec 1949	2.16	1.09	1.65	1.29	1.54	1.27
Jan 1950–Dec 1959	2.48	1.48	1.95	1.38	1.55	1.27
Jan 1960–Dec 1969	1.81	0.31	1.31	0.68	1.01	0.42
Jan 1970–Dec 1979	2.24	0.82	2.13	1.34	1.91	1.66
Jan 1980–Dec 1989	1.77	0.40	1.09	0.50	1.46	1.07
Jan 1990–Dec 1999	1.13	0.49	1.52	0.75	1.38	1.20
Jan 2000–Dec 2009	0.70	0.38	0.67	0.66	1.10	0.86
Jan 2010–Dec 2016	0.06	0.13	0.30	0.33	0.73	0.70

Notes: This exhibit shows the annualized gross Sharpe ratio (i.e., excess return before simulated transaction costs and fees divided by volatility) separately for each time-series momentum signal based on the past 1-month, 3-month, and 12-month trend, respectively. Also, the table shows the performance when each of these signals is lagged by one month.

간단한 관리 선물 전략의 백테스트: 12개월

- Moskowitz, Ooi and Pedersen (2012, JFE)



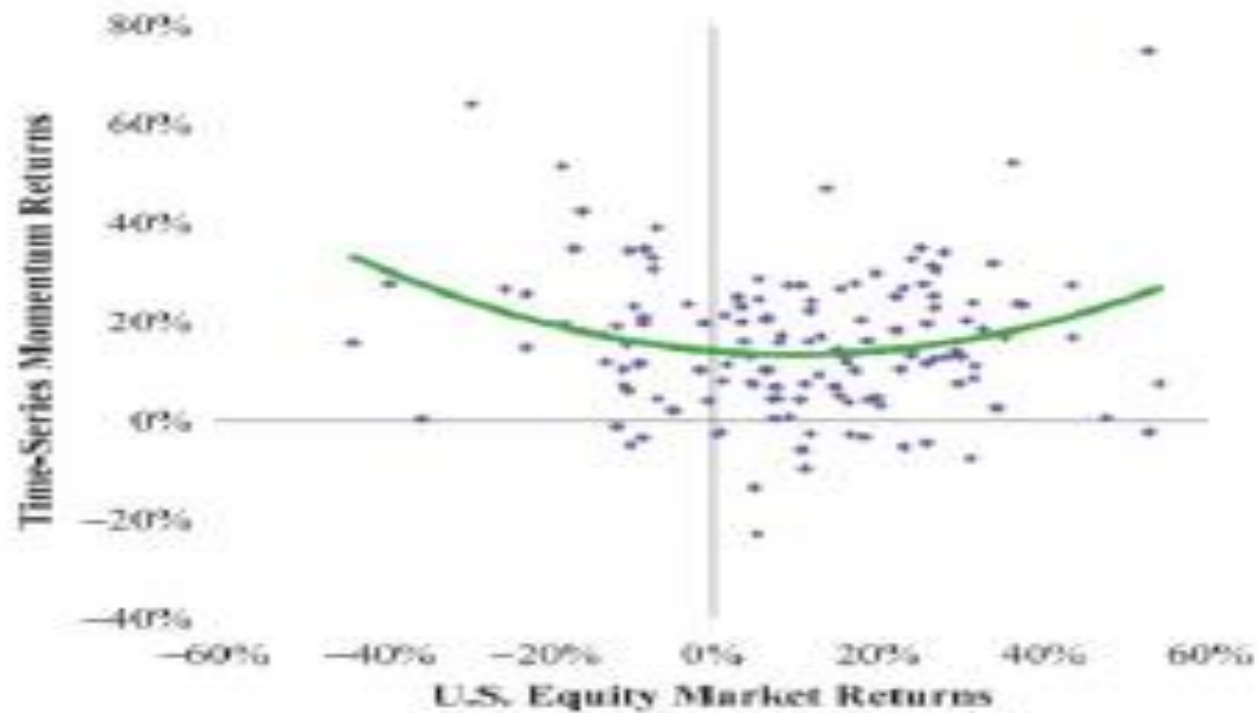
Fig. 4. The time series momentum smile. The non-overlapping quarterly returns on the diversified (equally weighted across all contracts) 12-month time series momentum or trend strategy are plotted against the contemporaneous returns on the S&P 500.

간단한 관리선물 전략의 백테스트: 100년의 증거

- Hurst, Ooi and Pedersen (2017, JPM)

EXHIBIT 5

Time-Series Momentum “Smile,” 1880–2016



Note: This exhibit shows the annual returns of time-series momentum (gross of fee, net of cost) vs. U.S. equity market returns, as well as the fitted second-order polynomial.

간단한 관리 선물 거래를 포트폴리오로 결합

- Moskowitz, Ooi and Pedersen (2012, JFE)

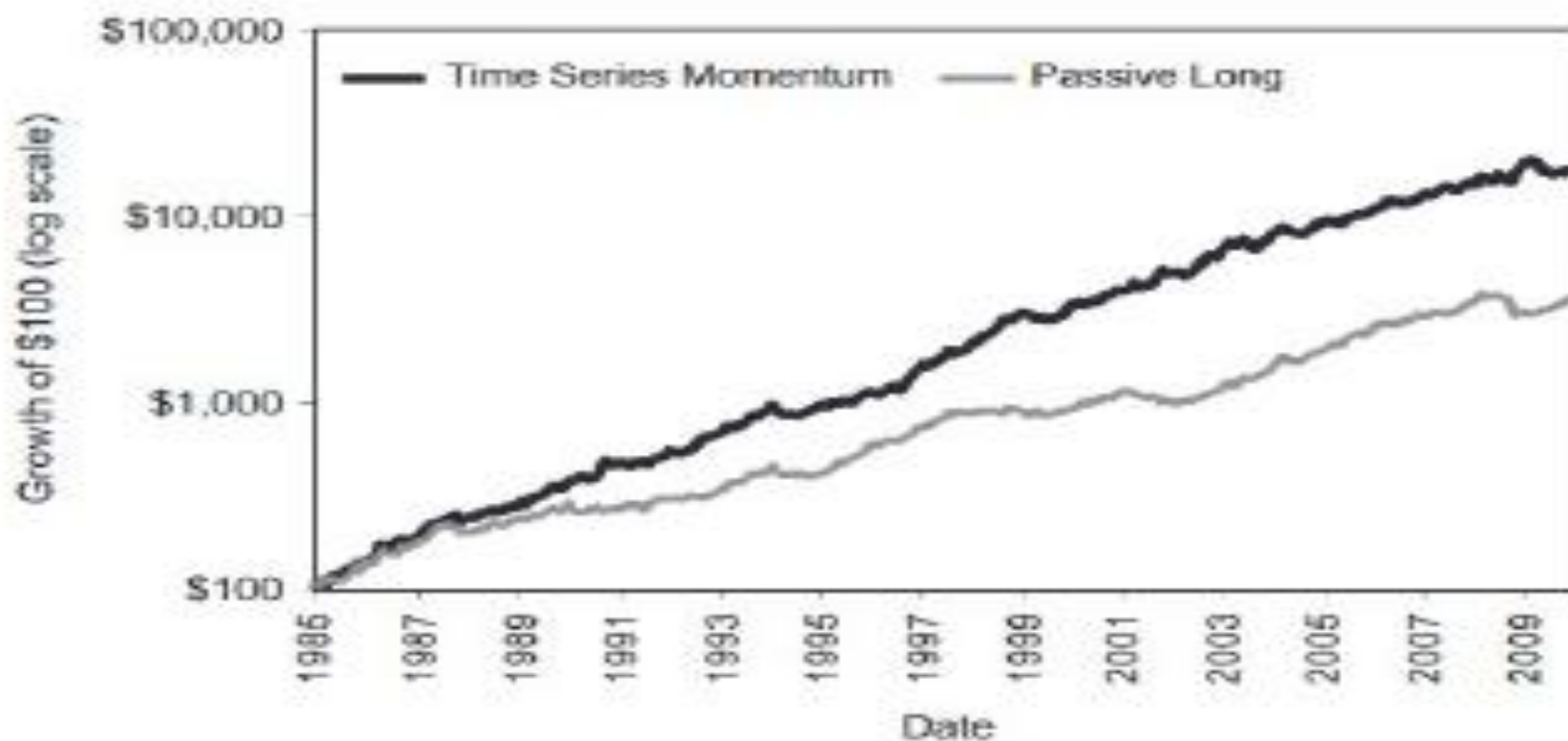


Fig. 3. Cumulative excess return of time series momentum and diversified passive long strategy, January 1985 to December 2009. Plotted are the cumulative excess returns of the diversified TSMOM portfolio and a diversified portfolio of the possible long position in every futures contract we study. The TSMOM portfolio is defined in Eq. (5) and across all futures contracts summed. Sample period is January 1985 to December 2009.

구현, 비용과 위험

- 관리 선물은 상당한 거래를 필요로 하므로 거래 비용을 관리해야 한다.
- 변동성이 큰 시장과 추세가 반전될 때 손실이 발생하므로 위험 관리는 중요하다.
- 전략이 여전히 유효한지 판단하기 어렵고, 손실이 더 나은 미래 투자 기회를 나타내는 것은 아니다 (가치 투자와 비교할 때).